MASHUK ELAHI

Portfolio Outline — Mission-Critical Power Systems & Critical Facilities

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Six case studies below, each reframed around technical content rather than original industry branding, using the Level 1–6 commissioning framework. Two pieces specifically showcase the strongest differentiators: Python EPMS automation (#5) and IEC 61850 / ABB 800xA protection-SCADA depth (#1, #2). The same engineering discipline — N+1/2N+1 redundancy, zero-tolerance commissioning, first-pass certification — applies whether the environment is a semiconductor fab, a heavy-industry power plant, or a Tier III/IV critical facility.

CASE STUDIES

1. MV 11kV Switchgear & ABB 800xA SCADA Fire & Gas Protection Commissioning

Challenge: Energise a large-scale heavy-industry MV/LV power system and SCADA-based fire & gas protection system with zero punch-list items at handover, coordinating crews of 100+.

My Commissioning Role (L1–L5): Level 1 FAT review of switchgear and protection panels; Level 2 installation oversight; Level 3 Energisation of MV 11kV switchgear; Level 4 FPT of protection-relay coordination and SCADA logic; Level 5 IST of integrated fire & gas protection.

Systems: MV 11kV switchgear, ABB 800xA SCADA, fire & gas protection logic.

Tools: Protection relay test sets, SCADA configuration tools, cause-and-effect matrices.

Quantified Result: 100% MV/LV uptime through turnover; zero outstanding punch-list items at handover.

2. IEC 61850 Protection-Relay Commissioning for a Wind-Power Substation

Challenge: Bring a wind-power substation electrical system to full IEC 61850 protection-relay conformity ahead of the energisation deadline.

My Commissioning Role (L1–L5): Level 3 Energisation; Level 4 FPT (protection-relay coordination, alarm/SCADA validation); Level 5 IST sign-off.

Systems: IEC 61850 protection relays, SCADA alarm/monitoring.

Tools: Relay test sets, IEC 61850 configuration/communication tools.

Quantified Result: 100% IEC 61850 conformity; zero punch-list items outstanding at handover.

3. 300mm Semiconductor Cleanroom Critical-Power Hook-Up & Functional Performance Testing

Challenge: Energise critical power for a 300mm semiconductor cleanroom with zero tolerance for downtime or contamination, to IEC 60204/61511/ISO 14644.

My Commissioning Role (L1–L5): Level 3 Energisation/Systems Start-up; Level 4 FPT of PLC/SCADA interlocks and functional safety.

Systems: PLC/SCADA interlocks, critical power distribution, cleanroom EPMS.

Tools: Prefabricated power skids, functional safety validation test plans.

Quantified Result: Zero audit rework; 100% tool-uptime/N+1 availability targets met; ~50% reduction in on-site installation labour via prefabrication — directly transferable to critical-facility white-space fit-out.

4. 72kWh BESS & EPMS Integration for a Hybrid Power Architecture

Challenge: Integrate a 72kWh Li-ion Battery Energy Storage System into an existing power architecture to cut peak energy draw without compromising reliability.

My Commissioning Role (L1–L5): Level 2 Installation; Level 3 Energisation / BMS-to-EPMS interface integration; Level 4 FAT acceptance.

Systems: BESS, BMS, DC/DC inverters, EPMS interface.

Tools: BMS configuration software, FAT test protocols.

Quantified Result: Up to 50% reduction in grid/fuel energy draw during low-load operation; zero rework at FAT.

5. Python-Automated EPMS Load-List & Cable-Schedule Generation at Scale

Challenge: Manually building and cross-verifying load lists and cable schedules across a multi-site power programme was slow and error-prone.

My Commissioning Role (L1–L5): Designed and deployed Python scripts to auto-generate and cross-verify load lists and cable schedules against EPMS/point-to-point data, flagging discrepancies automatically.

Systems: EPMS data exports, point-to-point verification logic.

Tools: Python.

Quantified Result: First-pass approval across four independent certification bodies with zero rework cycles; approach scales to large critical-facility EPMS point counts.

6. Autonomous Remote Power Installation — Solar-Wind Hybrid Commissioning

Challenge: Commission Malaysia’s first autonomous, unmanned solar-wind remote power installation with a 24V DC solar/battery bus and zero on-site operator presence.

My Commissioning Role (L1–L5): Level 3 Energisation through Level 5 IST; remote-operation validation.

Systems: Solar-wind hybrid generation, 24V DC battery bus, remote monitoring/EPMS.

Tools: Remote monitoring/SCADA, battery management.

Quantified Result: Eliminated routine diesel generator use; zero unplanned shutdowns at commissioning; fully remote/unmanned operation enabled.